Isaac Kremer

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EDUCATION

Temple University — Expected Dec. 2028
B.S. in Electrical and Computer Engineering — Philadelphia, PA

Alvernia University — Aug. 2024 – Dec. 2025
Mechanical and Electrical Engineering Coursework, 4.0 GPA — Reading, PA

Relevant Coursework: Engineering Computation II; Digital Circuit Design; Circuits and Electronics II

EXPERIENCE

SFS Group — Jun. 2026 – Aug. 2026; Aug. 2025 – Dec. 2025
Engineering Intern — Reading, PA

- Conducted a sensor trade study for locating raised roof-cladding ribs beneath insulation, comparing pulse-induction, very-low-frequency (VLF), ground-penetrating radar (GPR), and magnetic sensing by detection range, cost, and signal consistency
- Selected pulse-induction sensing and adapted a Spirit PI ESP32 metal-detector design to publish estimated metal-to-coil distance through a web server
- Tested sensing-coil geometries and identified a rectangular coil as providing the clearest distinction between high and low roof ribs

Blue Origin — Jan. 2026 – May 2026
Electrical Systems Intern — Kent, WA

- Designed and released 8 hardware-in-the-loop (HIL) harnesses for the aft section of New Glenn Stage 2, using wiring interconnect diagram (WID) requirements for connectors, backshells, wire gauge, and cable diameter to route harnesses in Creo Parametric
- Derived harness lengths from CAD routing and generated engineering bills of materials (eBOMs) and WTParts in Windchill, supporting approximately 20% of the program cycle's harness change scope
- Migrated legacy harness data into a new WID-generation tool and debugged tool behavior to validate generated harness definitions

Kline Process Systems — Mar. 2024 – Aug. 2025
Co-op Electrical Technician — Reading, PA

- Installed and wired industrial process-automation systems, including installation work at SpaceX, across control panels, pressure, temperature, level, and flow instrumentation, motors, solenoid valves, and variable-frequency drives (VFDs)
- Translated piping and instrumentation diagrams (P&IDs), panel drawings, and manufacturer schematics into field installations, independently determining conduit and cable routing
- Performed continuity testing, insulation testing, and I/O verification to validate wiring integrity and support system commissioning

PROJECTS

NASA L'SPACE Mission Concept Academy — Sep. 2025 – Dec. 2025
Electrical Engineer, Mission Concept Project — Remote

- Designed a 48 V rover electrical power subsystem integrating solar generation, battery energy storage, power conversion, and distribution within mission mass and power constraints
- Evaluated and selected GaAs solar arrays, $LiFePO_4$ batteries, maximum power point tracking (MPPT) controllers, and power-distribution hardware through performance, mass, cost, and mission trade studies
- Developed power-budget and battery state-of-charge (SOC) models to evaluate energy generation, consumption, and storage across mission operations

TECHNICAL SKILLS & CREDENTIALS

Programming & Systems: C, C++, Python, Bash, Linux/Unix, Git, GitHub
Embedded & Controls: ESP32, industrial automation, process automation, industrial instrumentation, control panels, VFDs, P&IDs, I/O verification
Test & Fabrication: Oscilloscope, signal generator, digital multimeter, soldering, continuity testing, insulation testing, tensile testing, 3D printing
CAD & PLM: Creo Parametric, Windchill, Fusion 360, AutoCAD, CATIA V5, Siemens NX